

From Income to Wealth Inequality

Trickle-Down vs. Capital Gains

Eustache Elina*

Raphaël Huleux[†]

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Abstract

An increase in permanent labor income inequality (PLI) increases aggregate savings because high-PLI households save more than poorer ones. For the asset market to clear, prices must adjust. This paper studies how this change in prices feeds back on the distribution of wealth. If higher households savings leads to more capital accumulation, the interest rate will decrease and wages will increase, "trickling-down" towards poorer households. If it only increases the price of financial assets, interest rates and wages will remain constant, and capital gains will increase the income of richer households. In a heterogeneous agent model with a non-homothetic taste for wealth and imperfect competition, we show that the level of markups dampens the trickle-down effect and increases the valuation effect. Our model calibrated to the U.S. economy suggests that the general-equilibrium effects of rising PLI inequality increased average wealth of the top 0.1% by 16% between 2020 and 1970, against 3% for the average household in the economy.

Keywords: Wealth inequality, permanent income, markups, capital gains, Tobin's Q

JEL codes: D31, E21, E25, E44

*Banque de France, eustache.elina@banque-france.fr

[†]University of Copenhagen, rph@econ.ku.dk

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Introduction

Permanent labor income inequality (PLI) has increased rapidly in the U.S. since the 1970s (DeBacker et al. 2011, Guvenen et al. 2021, 2022, Salgado et al. 2023, Braxton et al. 2025). Since households with higher permanent labor income have higher saving rates than poorer households, an increase in permanent labor income inequality increases aggregate savings¹. For markets to clear, the return on savings has to change, which has a feedback effect on the distribution of wealth. We ask: what is the sign and the magnitude of the general equilibrium effect of such an increase in aggregate savings on the distribution of wealth?

The answer to this question is theoretically ambiguous, and depends on the balance between two competing forces. The first is a *trickle-down channel*: an increase in aggregate savings can increase the quantity of physical capital, which will increase wages and decrease returns. This general equilibrium effect should dampen wealth inequality, as poor households benefit from higher wages, while rich households see their financial income decrease. The second is a *valuation channel*: an increase in aggregate savings can also increase the price of capital without expanding its quantity. This should increase wealth inequality: wages and long-term returns would remain constant, while short-term capital gains would temporarily increase the returns of the wealthy.

Market power determines which channel dominates. In a heterogeneous-agent model with permanent labor income heterogeneity, a non-homothetic taste for wealth, and Dixit-Stiglitz competition, we show that higher markups dampen the trickle-down channel while amplifying the valuation channel. Under imperfect competition, monopoly rents are discounted at lower rates, making firm valuations more sensitive to interest rate movements. When the interest rate decreases, the value of the firm increases faster than its stock of physical capital, increasing the average Tobin's Q. The net effect is that higher savings by the wealthy flow into asset prices rather than productive capital.

In an analytical two-agent version of our model, we prove that, at the steady state: (i) an increase in permanent labor income inequality decreases the interest rate, (ii) this general-equilibrium effect decreases wealth inequality, and (iii) markups dampen this trickle-down effect. Figure 1 illustrates this mechanism through a simple supply-demand diagram. An increase in PLI inequality shifts asset demand to the right. Under perfect com-

¹See Straub (2019), Auclert et al. (2025), among others.

petition, this expands physical capital substantially, lowering the interest rate and raising wages: the trickle-down channel. Under imperfect competition, the asset supply curve is flatter because it includes capitalized monopoly rents; the same demand shift produces a smaller decline in r and a smaller increase in K , but a larger increase in total asset value. The gap between ΔA and ΔK represents the rise in the average Tobin's Q —pure valuation gains that bypass the productive economy.

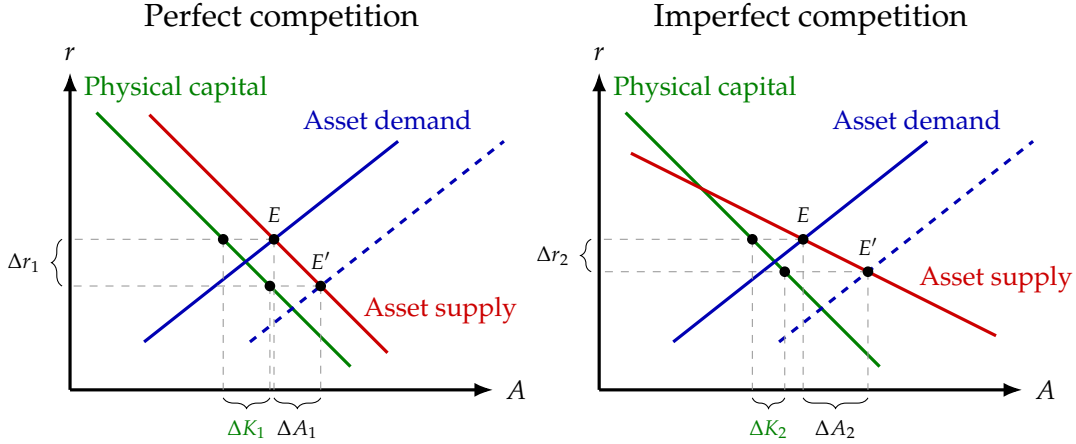


Figure 1: Long-term equilibria on the asset market following a PLI shock

While these steady-state results show that trickle-down prevails in the long run, the transition dynamics reveal a different picture. Because asset prices are forward-looking, the long-run decline in the interest rate is capitalized immediately into firm valuations, generating capital gains at the time of the shock. These capital gains accrue disproportionately to wealthier households who hold a larger share of risky assets, producing a fast and persistent rise in wealth inequality that precedes, and can dominate, the eventual trickle-down. Thus, while markups merely *dampen* the trickle-down channel at the steady state, along the transition they can *overturn* it: the general equilibrium effect switches sign, amplifying wealth inequality instead of reducing it in the short and medium run.

We quantify this mechanism by calibrating our model to the U.S. economy. Between 1970 and 2020, the general equilibrium effect of rising permanent labor income inequality increased average wealth of the top 1% by 10% and the top 0.1% by 16%, against 3% for an average household. Capital gains also have strong welfare implications: under perfect competition, the general-equilibrium effect decreases the welfare of the rich, whereas, under imperfect competition, the capital gains more than compensate the long-term decrease in returns, increasing the welfare of the top 1% and 0.1%.

Our model is also consistent with observed patterns of wealth accumulation in the U.S. It generates a long-term 70 p.p. increase in the wealth-to-output ratio and a 70 basis point decline in the real interest rate, with most of the wealth accumulation reflecting valuation effects rather than physical capital accumulation, as observed in the data. This aligns with the decline in the aggregate household net saving rate: households have accumulated wealth primarily through unrealized capital gains on existing assets, not by saving a larger share of their income.² Correspondingly, the value of the capital stock has risen substantially: average Tobin’s Q increased from around 0.6 in the 1960s to over 1 in recent decades (Brun & Gonzalez 2017, Gonzalez & Trivin 2019).

Related Literature. Our paper makes two key contributions. First, we show analytically that the general equilibrium effect of a rise in PLI inequality dampens wealth inequality at the steady state—a “trickle-down” effect—but that this effect weakens with markups. Second, we show quantitatively that along the transition, imperfect competition overturns the trickle-down: capital gains generated by higher firm valuations increase wealth inequality, and this effect dominates for decades.

Rising labor income inequality and the macroeconomy. The closest paper to ours is Straub (2019), who shows that non-homothetic preferences allow the increase in PLI inequality to explain a large part of the rise in wealth inequality and the decline in real interest rates. We differ on two dimensions: we focus on the general-equilibrium effect and show analytically how it dampens wealth inequality and improves welfare for the poor; and we introduce imperfect competition, which redirects higher savings into asset prices rather than productive capital. A large body of work studies the aggregate implications of rising labor income inequality (Heathcote et al. 2010, Favilukis 2013, Kumhof et al. 2015, Auclert et al. 2025). Relative to this literature, our contribution is to show that imperfect competition adds a valuation mechanism that further concentrates wealth at the top.

Our quantitative results on the decline in the interest rate are in line with the estimates in Straub (2019) and Auclert et al. (2025), and consistent with the macro-finance trends documented in Farhi & Gourio (2018). Our paper can be seen as providing a micro-foundation for the increase in savings supply that they generate through an exogenous shift in the discount factor.

Market power and inequality. A growing literature studies the link between market power

²Net saving rates exclude unrealized capital gains. Gross saving rates, by contrast, include them in both income (the Haig-Simons definition) and savings. National accounts measures of savings omit capital gains entirely.

and the distribution of income and wealth. Impullitti & Rendahl (2025) focuses on the link from market power to inequality, showing that higher barriers to entry raise markups and boost firm valuations, widening the return gap $r - g$ and amplifying wealth inequality. Liu et al. (2022) show that low interest rates may increase market concentration and reduce productivity growth. Moll et al. (2022) study how automation increases wealth inequality by raising returns to wealth, showing that these return movements, rather than labor income changes alone, are the key driver of uneven growth. Our paper shares the insight that return heterogeneity drives wealth concentration, but identifies a different source: endogenous capital gains from the interaction of non-homothetic savings with imperfect competition, rather than automation-driven changes in the marginal product of capital.

Dynamics of wealth inequality. Gabaix et al. (2016) show that most heterogeneous-agent models struggle to generate fast increases in wealth inequality, requiring either “type dependence” or “scale dependence.” We show that endogenous capital gains can generate the fast dynamics observed in the data, consistent with the empirical finding that richer households accumulate wealth primarily through capital gains (Fagereng et al. 2019) and earn higher returns (Fagereng et al. 2020, Bach et al. 2020, Garbinti et al. 2021). Cioffi (2025) builds a portfolio choice model with non-homothetic preferences where households shift from safe assets to housing and then to equity as wealth increases, showing that temporary equity return shocks have large and persistent effects on top wealth shares. Gomez (2025) shows that portfolio heterogeneity creates a feedback loop: equity returns raise top wealth shares, which in turn increase asset demand and prices. Our model features a related amplification—capital gains from higher firm valuations concentrate wealth at the top—but the source differs: in our setting, capital gains arise endogenously from the interaction of a PLI shock with imperfect competition, rather than from exogenous return shocks filtered through portfolio heterogeneity. Relative to Hubmer et al. (2021), who match the level and dynamics of wealth inequality using exogenous excess returns, our model endogenizes these heterogeneous returns through the interaction of non-homothetic savings and imperfect competition.

Outline. The remainder of the paper is organized as follows. Section 1 documents stylized facts on inequality and wealth accumulation. Section 2 presents the model. Section 3 derives analytical results in a two-agent economy. Section 4.1 calibrates the full quantitative model to the U.S. economy. Section 4 presents the main quantitative results. Section 5 concludes.

1 Stylized facts

This section documents the main stylized facts about the U.S. that motivate this paper. At the aggregate level, we report (i) the increase in labor income and wealth inequality since the 1970s, (ii) the role of capital gains in the accumulation of wealth. At the cross-sectional level, we use the Survey of Consumer Finances (SCF) to document that richer households (i) have a higher saving rate and (ii) invest a higher share of their wealth into risky assets.

Figures 2 and 3 report the main estimates of labor income and wealth inequality used in this paper, using data from Piketty et al. (2018), which combines tax, survey, and national accounts data to estimate the increase in the top labor and income shares. The primary advantage of their method is that it matches aggregate values of income and wealth in the national accounts. The values are similar, albeit a bit lower, to those obtained in the SCF.³ Both document a significant increase in the top 10% and top 1% in wealth and labor income shares since the 1970s.⁴

The distribution of wealth displays a fatter right tail than the distribution of labor income, and this tail has become even fatter over time. According to PSZ, the top 1% labor income share has risen by approximately 6 p.p. between 1980 and 2020, against approximately 13 p.p. for the top 1% wealth share. Moreover, the total wealth held by households increased faster than national income. It is important to note that because measures of national income do not account for capital gains, an increase in wealth does not necessarily come from increased net savings from households. Indeed, decomposing the wealth-to-output ratio between a saving and a capital gain component, the left panel of Figure 4 shows that almost all of the increase in the wealth-to-output ratio comes from an increase in the price of wealth and not from an increase in savings.⁵ This increase in the valuation of capital is also clearly visible in the valuation of the firms: the right panel of Figure 4 shows that the aggregate Tobin’s Q (the ratio of the market value and the book value of U.S. corporations)

³The PSZ distributional data use the equal-split adult as the unit of observation, while the SCF reports statistics at the household level. To ensure comparability, we apply an equal-split adjustment to the SCF: for married households, all dollar amounts are divided by two and survey weights are doubled. See Appendix A.2 for details.

⁴There is an active debate on the measurement of top income shares (Auten & Splinter 2024). This debate primarily concerns total income—especially the allocation of untaxed business and capital income—rather than labor income, which is largely observed in tax data. For wealth shares, Smith et al. (2023) offer revised estimates that differ from PSZ in level but confirm the qualitative rise. We use PSZ as our primary source because it ensures consistency with national accounts aggregates.

⁵That is, $\frac{W_t}{GNP_t} = \frac{W_{t-1} + S_{t-1} + KG_{t-1}}{GNP_t}$, where W_t is the total real wealth held by households, S_t is total savings and KG_t are capital gains.

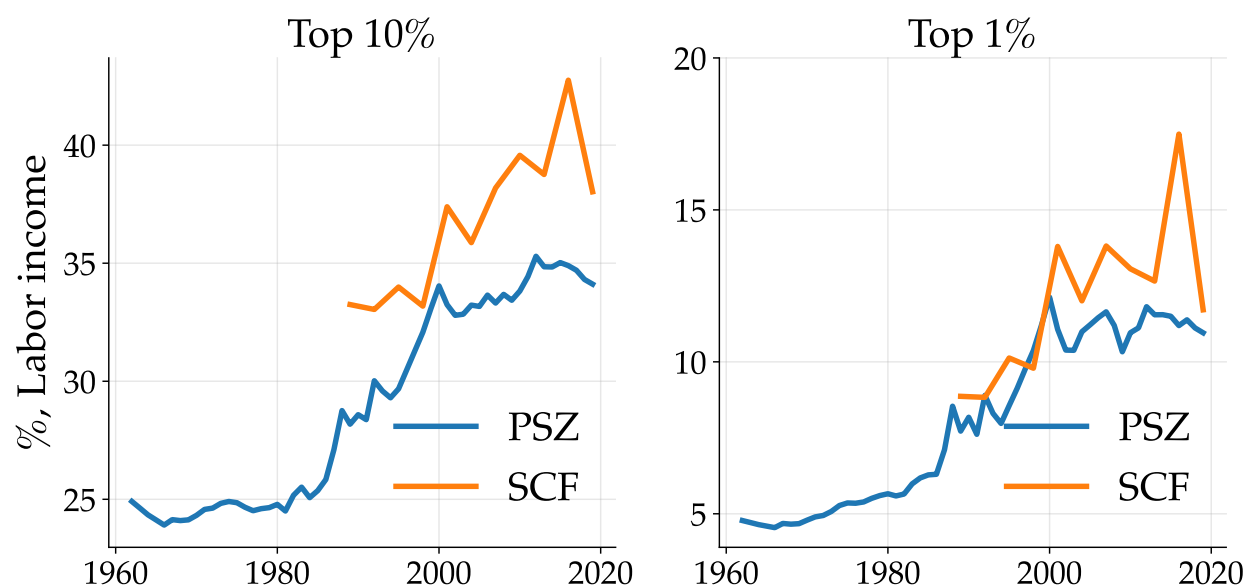


Figure 2: Top labor income shares

Note: This figure shows the evolution of the top 1% and top 10% pre-tax labor income shares from the Distributional National Accounts in Piketty et al. (2018) (PSZ, blue line) and in the Survey of Consumer Finances (SCF, orange line, authors calculations).

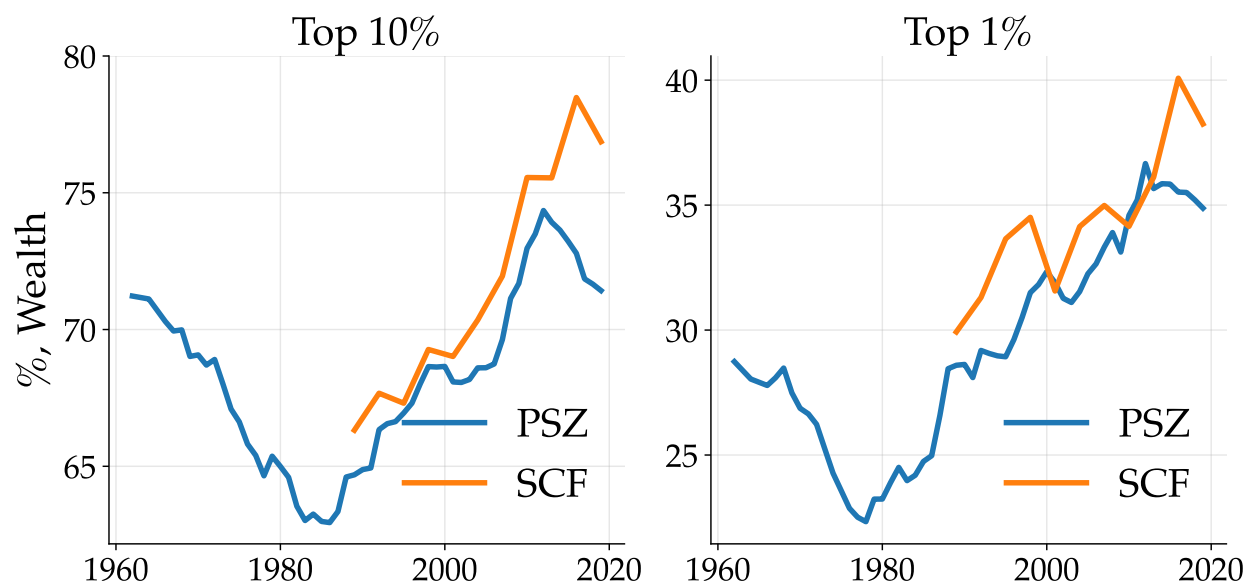


Figure 3: Top wealth shares

Note: This figure shows the evolution of the top 1% and top 10% wealth shares from the Distributional National Accounts in Piketty et al. (2018) (PSZ, blue line) and in the Survey of Consumer Finances (SCF, orange line, authors calculations).

went from below 1 before the 1980s to above 1 today.

We note two important points. First, this decomposition is not a counterfactual exercise: in the absence of capital gains, households might have increased their savings rate to satisfy their savings needs. It is, however, consistent with Fagereng et al. (2019), who found the higher saving rates from richer households come mostly from capital gains. Secondly, this increase in the valuation of the firms is also due to other factors than the increase in labor income inequality: the decrease of the dividend and the corporate tax rate and the increase in markups also played an important role (Piketty & Zucman 2014, Brun & Gonzalez 2017). This considerable accumulation of wealth at the aggregate level has also been accompanied by an increase in the equity premium (Caballero et al. 2017, Reis 2022).

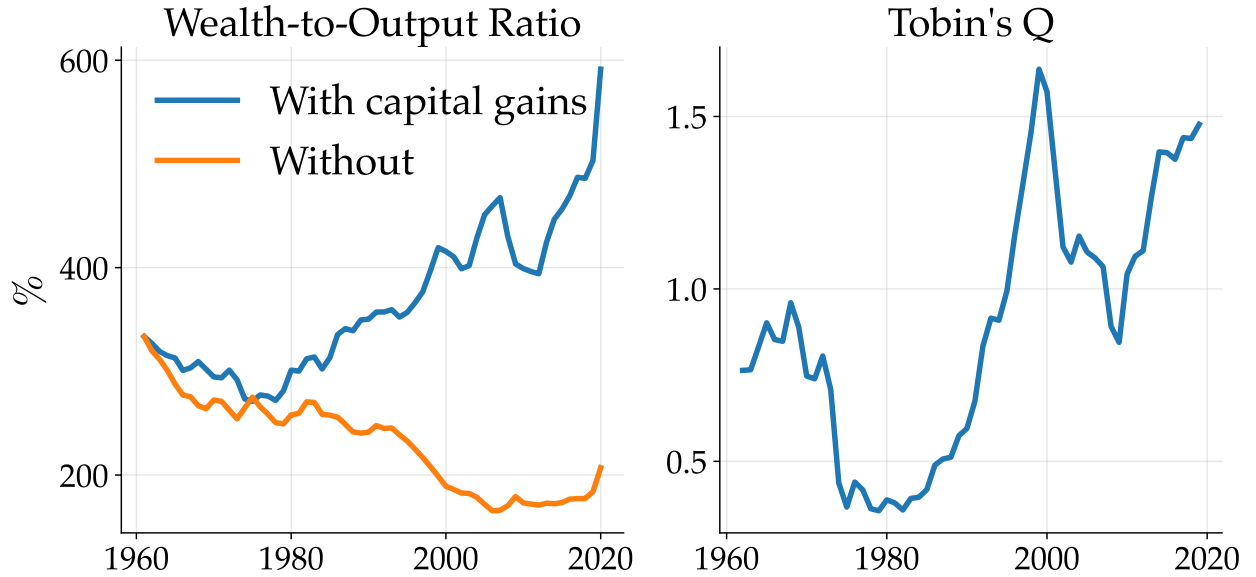


Figure 4: Wealth-to-output ratio and Tobin's Q

Note: The left panel shows the evolution of the household wealth-to-national-output ratio. The decomposition follows Piketty & Zucman (2014) by computing aggregate wealth removing any valuation effect, using the law of motion $W_{t+1} = W_t + S_t + KG_t$ where W_t is the household aggregate wealth, S_t is aggregate savings from households and KG_t are capital gains. The right panel shows the aggregate Tobin's Q reported in Piketty et al. (2018).

Figure 5 highlights two key stylized facts at the cross-sectional level: (i) the saving rate is an increasing function of permanent income, and (ii) richer households invest a higher share of their wealth into risky assets.

Those two facts are crucial to explaining the primary mechanism in our model: the difference in saving rates along the distribution of permanent income implies that an increase in labor income inequality can affect the desired aggregate savings. Those changes will, in turn, affect the valuation of capital, which will create an excess return on capital com-

pared to the safe asset. Because richer households own a higher share of risky assets, this effect will amplify wealth inequality, creating a feedback loop between labor income and wealth inequality.

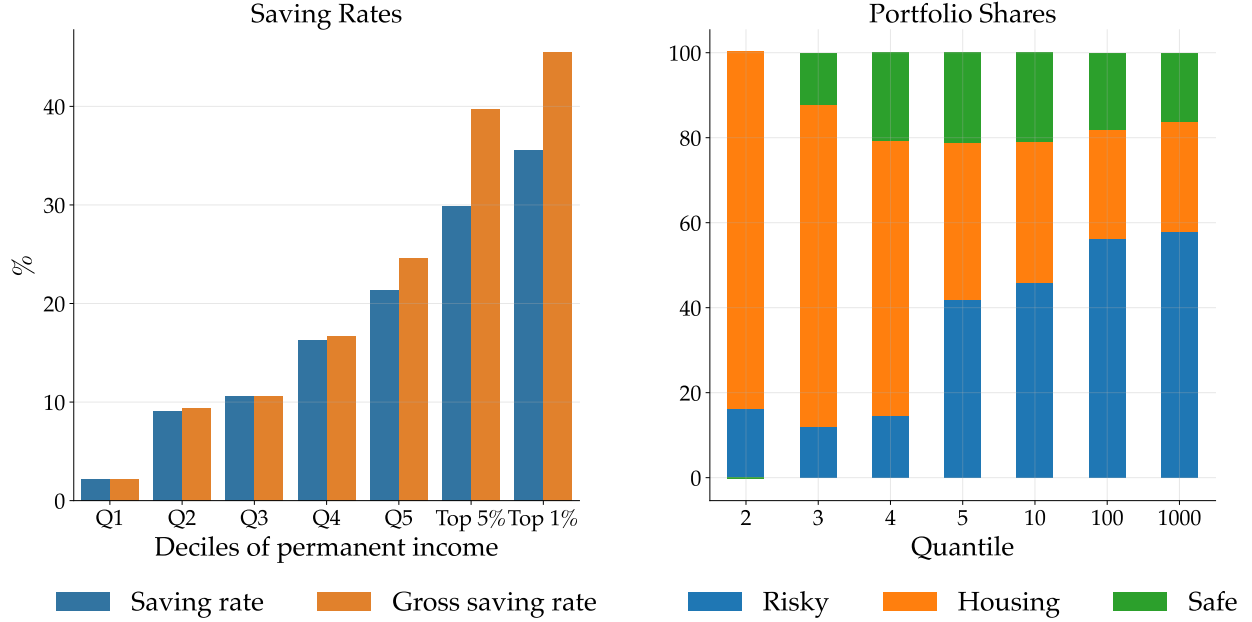


Figure 5: Saving rates by permanent income and portfolio shares by wealth in the SCF, 1989

Note: The left panel reproduces average saving rates by decile of permanent income in the SCF in 1989, as computed by Kumhof et al. (2015) using a quantile regression controlling for age. Gross saving rates account for unrealized capital gains. The right panel shows portfolio composition by wealth percentile in the SCF in 1989 (authors' calculations). Risky assets include equities and mutual funds; safe assets include deposits and bonds; housing includes the net value of primary and secondary residences.

2 Model

The economy is populated by a continuum of households, a continuum of intermediate goods producing firms, a representative final good producer, a mutual fund, and a government. Time is discrete and denoted t . There is no aggregate risk, and changes to the distribution of permanent labor income are modeled as MIT shocks.⁶

⁶An MIT shock is a one-time unexpected shock on some parameter. From the model point of view, this shock is a zero-probability event, and the following transition towards the steady state is completely deterministic. Thus households do not have rational expectations with respect to the variable that is shocked, in this case, the distribution of permanent income.

2.1 Households

Households save in a mutual fund, subject to a zero borrowing-constraint, and gain labor income at wage w_t , subject to a progressive tax à la Feldstein (1969), B  nabou (2002), and Heathcote et al. (2017), with progressivity parameter λ and level τ . They differ in their level of permanent income z_i , and are subject to uninsurable idiosyncratic productivity shocks e_t . They discount the future at a rate β , face a constant death probability ξ , and derive utility from consumption c_t and from their total wealth holdings a_t . Dying households are reborn with zero wealth, to ensure a constant mass of households, and conserve their permanent productivity type. Their maximization problem is

$$\begin{aligned} V_t(a_{t-1}, e_t, z_i) &= \max_{a_t} \left\{ u(c_t) + v(a_t) + (1 - \xi)\beta \mathbb{E}_e[V_{t+1}(a_t, e_{t+1}, z_i)] \right\}, \\ \text{s.t. } c_t + a_t &= \underbrace{(1 + r_t)a_{t-1}}_{\text{financial income}} + \underbrace{s(a_{t-1}, A_{t-1})cap_t}_{\text{capital gains}} + \underbrace{(1 - \tau)(z_i e_t w_t)^{1-\lambda}}_{\text{post-tax labor income}} + \underbrace{z_i \eta_t}_{\text{bequests}}, \quad a_t \geq 0. \end{aligned} \quad (1)$$

In Equation (2), r_t is the interest rate, cap_t denotes the capital gains (to be defined below), A_{t-1} the total wealth in the economy, and $s(a_{t-1}, A_{t-1})$ is the household's exposure to equity, which is assumed to follow an exogenous, wealth-dependent rule:

$$s(a_{t-1}, A_{t-1}) = \frac{\psi_0}{1 + \exp(-\psi_1(a_{t-1}/A_{t-1} - \psi_2))} a_{t-1}. \quad (3)$$

This reduced-form rule captures the well-documented heterogeneity in portfolio composition along the wealth distribution—wealthier households hold a larger share of their portfolio in risky assets—without requiring a full portfolio choice problem.⁷

All households supply exogenously a single unit of labor, with productivity $e_t z_i$, where the idiosyncratic productivity e_t follows a log-AR(1) process:

$$\log e_t = \rho \log e_{t-1} + \varepsilon_t^e, \quad \varepsilon_t^e \sim \mathcal{N}(0, \sigma^e), \quad (4)$$

which is discretized using the Rouwenhorst method. The total endowment of labor productivity $\int z_i e_t d\Phi_t$ is normalized to 1, where Φ_t denotes the joint distribution over wealth, idiosyncratic productivity, and permanent productivity. Through the lens of our model, an increase in permanent labor income inequality is a mean-preserving spread in the dis-

⁷See Appendix A.3 for the calibration of (ψ_0, ψ_1, ψ_2) to the Survey of Consumer Finance, following Auclert & Rognlie (2020).

tribution of z_i such that $\int_i z_i d\Phi_t = 1$.

η_t is the total amount of accidental bequests in the economy left by dying households, which we assume are paid to living households proportional to their permanent productivity. The accidental bequests are defined as

$$\eta_t = \xi \left(A_{t-1}(1 + r_t) + cap_t \int s(a_{t-1}, A_{t-1}) d\Phi_t \right). \quad (5)$$

Period utility functions are given by

$$u(c_t) = \frac{c_t^{1-\sigma}}{1-\sigma}, \quad v(a_t) = \gamma \frac{(a_t + \zeta)^{1-\Sigma}}{1-\Sigma},$$

where σ is the coefficient of relative risk aversion, γ the intensity of the taste for wealth, ζ a Stone-Geary shifter, and Σ the curvature of wealth in utility. We assume $\Sigma < \sigma$, so that the marginal propensity to save increases with permanent income.

2.2 Government

The budget constraint of the government is

$$B_t = B_{t-1}(1 + r_t) + G_t - T_t$$

where G_t denotes government spending, T_t total tax receipts, and B_t real government bonds. The government finances its outlays by imposing a progressive tax on labor income and taxing dividends d_t at rate τ :⁸

$$T_t = \tau d_t + L_t w_t - (1 - \tau) \int (w_t e_t z_i)^{1-\lambda} d\Phi_t.$$

We assume a balanced-budget rule that keeps public debt constant at $B_t = B$ and government spending constant at $G_t = G$, so that the budget constraint simplifies to:

$$G = T_t - Br_t.$$

⁸The term $Lw_t - (1 - \tau) \int (w_t e_t z_i)^{1-\lambda} d\Phi_t$ can be interpreted as the government taxing all labor income in the economy, and then redistributing an amount $(1 - \tau) \int (w_t e_t z_i)^{1-\lambda} d\Phi_t$ to households.

2.3 Firms

The supply side features monopolistic competition in the intermediate-goods sector and perfect competition in the final-goods sector.

Final-good producer. A competitive firm produces the final good (the numéraire) by combining labor L_t and a CES bundle of intermediate varieties $j \in [0, 1]$:

$$Y_t = Z L_t^{1-\alpha} X_t^\alpha, \quad X_t \equiv \left(\int_0^1 x_{j,t}^{\frac{1}{\mu}} dj \right)^\mu,$$

where Z is total factor productivity, α the capital share, and $\mu \geq 1$ is the markup, which parameterizes the substitutability across varieties.⁹ The final-good producer chooses labor and intermediate inputs to maximize profits:

$$\max_{L_t, \{x_{j,t}\}} Y_t - w_t L_t - \int_0^1 p_{j,t} x_{j,t} dj.$$

The first-order conditions are

$$w_t = (1 - \alpha) \frac{Y_t}{L_t},$$

$$p_{j,t} = \alpha \frac{Y_t}{X_t} \left(\frac{X_t}{x_{j,t}} \right)^{\frac{\mu-1}{\mu}}.$$

Defining $P_t^X \equiv \alpha Y_t / X_t$ as the price of the intermediate aggregate, the second condition can be inverted to obtain the demand for variety j :

$$x_{j,t} = X_t \left(\frac{p_{j,t}}{P_t^X} \right)^{-\frac{\mu}{\mu-1}}. \quad (6)$$

Intermediate producer. Each intermediate firm j owns its capital stock $k_{j,t-1}$ (predetermined), operates the linear technology $x_{j,t} = k_{j,t-1}$, and pays a per-period fixed operating cost $F \geq 0$ (in units of the final good). Taking the demand curve from Equation (6) as given, firm j chooses its price $p_{j,t}$ and investment $k_{j,t}$ to maximize the cum-dividend value of the firm $d_{j,t} + q_{j,t}$, where dividends are

$$d_{j,t} = p_{j,t} x_{j,t}(p_{j,t}) - [k_{j,t} - (1 - \delta) k_{j,t-1}] - F,$$

⁹The elasticity of substitution across varieties is $\varepsilon = \mu / (\mu - 1)$, so that μ is the equilibrium markup of each intermediate producer.

with δ the depreciation rate of capital, and the end-of-period equity value $q_{j,t}$ satisfies the no-arbitrage condition

$$(1 + r_t) q_{j,t-1} = (1 - \tau) d_{j,t} + q_{j,t}. \quad (7)$$

The constant-elasticity demand implies the standard markup pricing rule $p_{j,t} = \mu mc_{j,t}$, where the marginal cost $mc_{j,t}$ equals the user cost of capital $r_t + \delta$, obtained from the first-order condition for investment.¹⁰ Hence $p_{j,t} = \mu (r_t + \delta)$.

Symmetric equilibrium. Since all intermediate firms face identical costs and demand conditions, the equilibrium is symmetric: $x_{j,t} = x_t$ and $p_{j,t} = p_t$ for all j . The CES aggregator then gives

$$X_t = \left(\int_0^1 x_t^{\frac{1}{\mu}} dj \right)^\mu = x_t,$$

so that $X_t = K_{t-1}$ in the aggregate, where K_{t-1} is the predetermined capital stock.¹¹ Combining with the markup pricing rule $p_t = \mu (r_t + \delta)$ yields

$$r_t = \frac{\alpha}{\mu} \frac{Y_t}{K_{t-1}} - \delta. \quad (8)$$

Imposing equilibrium on the labor market $L_t = L = 1$, total output is $Y_t = Z K_{t-1}^\alpha$. Aggregate dividends are

$$d_t = \alpha Y_t - K_t + (1 - \delta)K_{t-1} - F.$$

2.4 Mutual fund

A mutual fund receives all savings from households a_t and invests them in firm equity q_t and government bonds B . We assume that firm equity q_t is in unit net supply. The no-arbitrage condition (7) ensures that the return on equity equals r_t . Combined with asset market clearing and zero profits in the fund market, this implies that the return on households' assets also equals r_t .

However, in the first period after an unexpected shock, the no-arbitrage condition (7) does

¹⁰The first-order condition for $k_{j,t}$ equates the marginal cost of investment (one unit of the final good) to the discounted marginal return: $1 = (p_{j,t+1} + 1 - \delta) / (1 + r_{t+1})$, yielding $p_{j,t} = r_t + \delta$.

¹¹In symmetric equilibrium $p_t = P_t^X = \alpha Y_t / K_{t-1}$. Since the final good is the numéraire, P_t^X is an endogenous price that does not require a separate normalization; it is eliminated by combining the final-good and intermediate FOCs.

not hold, since the jump in q_0 and d_0 is unanticipated and was not priced into $q_{-1} = q_{ss}$. The unexpected excess return is paid to households according to the portfolio rule from Equation (3), and the balance sheet of the mutual fund at $t = 0$ writes

$$A_{ss}(1 + r_{ss}) + cap_0 \int s(a_{ss}, A_{ss}) d\Phi_{ss} = q_0 + (1 - \tau)d_0 + B(1 + r_{ss}),$$

which implies that capital gains in period 0 are equal to

$$cap_0 = \frac{q_0 - q_{ss} + (1 - \tau)(d_0 - d_{ss})}{\int s(a_{ss}, A_{ss}) d\Phi_{ss}}.$$

For tractability, we also define a return including capital gains

$$r_t^{full} = r_t + cap_t \frac{A_{t-1}}{\int s(a_{t-1}, A_{t-1}) d\Phi_{t-1}}.$$

2.5 Market Clearing and Equilibrium

Definition 1 (Equilibrium). *Let $\mathcal{A}$ be the space for assets, $\mathcal{E}$ the space for productivity, and $\mathcal{Z}$ the space for permanent type.*

An equilibrium in this economy is defined as paths for household and firm decisions $\{a_t, c_t, K_t, d_t\}_{t=0}^\infty$, prices $\{r_t, q_t, w_t\}_{t=0}^\infty$, aggregate quantities $\{\eta_t\}_{t=0}^\infty$, capital gains cap_0 , the tax policy $\{\lambda_t, \tau\}_{t=0}^\infty$, and distributions $\{\Phi_t\}_{t=0}^\infty$, such that, for all t : (1) households and firms solve their objective functions, (2) the sequence of distributions satisfies aggregate consistency conditions, (3) the government budget holds, and (4) all markets clear at all times:

1. *The asset market clears: $\int_{\mathcal{A} \times \mathcal{E} \times \mathcal{Z}} a_t d\Phi_t = B + q_t$.*
2. *The labor market clears: $L_t = 1$.*
3. *The goods market clears: $Y_t = C_t + K_t - (1 - \delta)K_{t-1} + G$.*

3 Analytical Results

This paper examines the general-equilibrium effects of rising permanent labor income inequality. To build intuition, we begin with a simplified two-agent version of the model

presented in Section 2. We analyze a mean-preserving spread in which the rich agent's labor income z_1 increases while the poor agent's income $z_2 = 1 - z_1$ decreases by an equivalent amount.

We also impose the following simplifications: the taste-for-wealth coefficient is normalized to $\gamma = 1$, a zero Stone-Geary constant ($\zeta = 0$), no idiosyncratic risk, death, or bequests, no government sector (zero taxes, public debt, and spending), and zero fixed costs F .¹² We assume a steady state exists and compare steady states before and after the increase in permanent labor income inequality.¹³

We establish three results. First, an increase in permanent labor income inequality lowers the interest rate and raises the wealth-to-output ratio. Second, the general-equilibrium response dampens the resulting increase in wealth inequality. Third, this “trickle-down” effect weakens with markups.

3.1 General-Equilibrium Effect of a Rise in Permanent Labor Income Inequality

Throughout this section, we focus on the case where wealth is a luxury good: $\sigma > \Sigma$. We further maintain the following assumption.

Assumption 1. *The share of capital income in consumption of the richest agent satisfies $ra_1/c_1 < \Sigma/\sigma$.*

Proposition 1. *Under Assumption 1, an increase in permanent labor income inequality increases desired savings in partial equilibrium:*

$$\partial \log A \equiv \sum_i \frac{\partial \log(A^i)}{\partial z_i} dz_i > 0. \quad (9)$$

Furthermore, the semi-elasticities of asset demand and supply $\varepsilon^A \equiv \partial \log A / \partial r$ and $\varepsilon^q \equiv -\partial \log q / \partial r$ are strictly positive, so that, for a given $\partial \log A$:

1. The equilibrium interest rate decreases

$$dr = -\frac{\partial \log A}{\varepsilon^A + \varepsilon^q} < 0, \quad (10)$$

¹²The $\zeta = 0$ assumption insures that the borrowing constraint will never be binding for any agent. The $F = 0$ assumption simplifies the algebra without loss of generality.

¹³Additional details are provided in Online Appendix A.1.

2. *The equilibrium wage increases*

$$d \log w = -\alpha \varepsilon^K dr > 0, \quad (11)$$

3. *The equilibrium wealth-to-output ratio increases*

$$d \log \frac{q}{Y} = \frac{\varepsilon^q - \alpha \varepsilon^K}{\varepsilon^A + \varepsilon^q} \partial \log A > 0. \quad (12)$$

where $\varepsilon^K \equiv -d \log(K)/dr$ is the semi-elasticity of the capital stock with respect to the interest rate.

Proof. See Online Appendix A.2. □

Equations 10 and 12 in Proposition 1 are the counterparts of the decomposition in Auclert et al. (2025) in our setting. The impact of higher permanent labor income inequality on the interest rate follows a standard supply-and-demand logic: when asset demand rises, prices must adjust to clear markets. This can be seen in Figure 6: the increase in permanent labor income inequality shifts the asset-demand curve to the right, raising the desired aggregate savings A at a given interest rate r . The equilibrium is determined by the intersection of the asset-supply (the price of the firm q) and desired savings A . The equilibrium change dr depends on the semi-elasticities of asset supply and demand, ε^q and ε^A .

The intuition behind Proposition 1 is as follows. When $\Sigma < \sigma$, the taste for wealth is non-homothetic: wealth acts as a luxury good. Households with higher permanent labor income hold proportionally more wealth for the social status it confers, making savings an increasing and convex function of permanent income, as we documented in Section 1. A mean-preserving spread in permanent labor income therefore raises desired aggregate savings, $d \log A > 0$. Since both semi-elasticities are positive, $dr < 0$. The decline in the interest rate raises the value of firms and, in turn, the wealth-to-output ratio.

In the absence of a non-homothetic taste for wealth (either because $\gamma = 0$ or because $\Sigma = \sigma$), savings would be linear in permanent labor income and the increase in permanent labor income inequality would not raise desired savings, so $\partial \log A = 0$. In this case, there would be no change in the interest rate, and the wealth-to-output ratio would remain unchanged. This highlights the critical role of non-homothetic wealth preferences

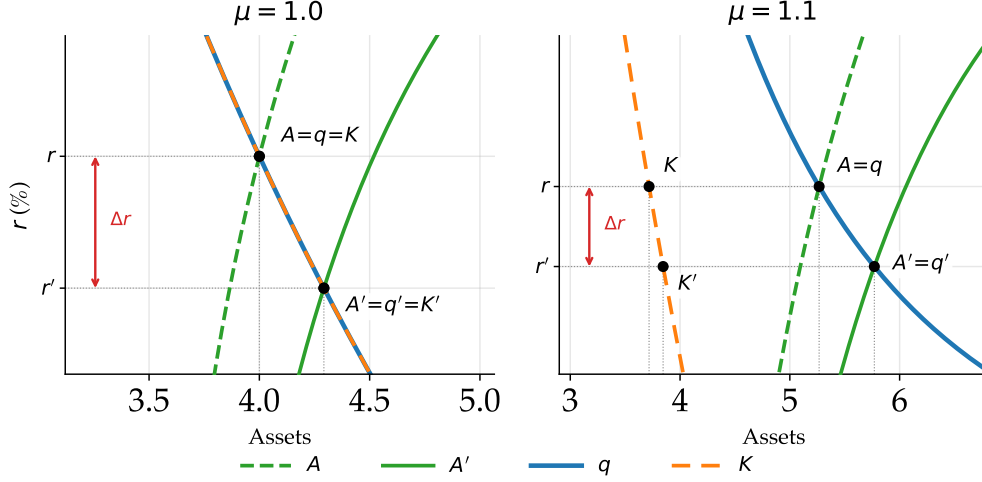


Figure 6: Impact of a rise in permanent labor income inequality on the supply and demand for assets

Note: The left panel shows the equilibrium on the asset market at the steady-state, under perfect competition. The equilibrium is determined by the intersection of the asset-supply (the price of the firm q) and desired savings A . Under imperfect competition, the asset-supply curve is steeper, and the demand for physical capital differs from the firm's valuation.

in both matching the observed relationship between savings and permanent income, and generating the general-equilibrium effects described in Proposition 1.

Finally, we verify that Assumption 1 holds numerically. This is generally the case, and it holds in particular for $\Sigma/\sigma = 0.445$, our baseline calibration.

3.2 The “Trickle-Down” Effect

We established that a rise in permanent labor income inequality lowers the interest rate. The resulting decline in r raises the price of the firm q and the capital stock K , altering the endogenous prices that govern household wealth accumulation: the return on assets r and wages w . These price adjustments, in turn, affect the distribution of wealth—the central object of interest in this paper.

Proposition 2. Define $\varepsilon^{a_i} \equiv d \log(a_i) / dr$ for $i = 1, 2$ the semi-elasticity of the wealth of agent i . Under Assumption 1, the non-homothetic taste for wealth implies that the semi-elasticity of wealth is higher for the rich than for the poor, $\varepsilon^{a_1} > \varepsilon^{a_2} > 0$, so that:

$$\varepsilon^{a_1} - \varepsilon^{a_2} > 0, \quad (13)$$

and the general equilibrium effect ($dr < 0$) dampens the increase in wealth inequality following an

increase in permanent labor income inequality:

$$(\varepsilon^{a_1} - \varepsilon^A) dr < 0. \quad (14)$$

Proof. See Online Appendix A.3. □

The key insight of Proposition 2 is that non-homothetic wealth preferences imply $\varepsilon^{a_1} > \varepsilon^{a_2}$: the wealth of the rich is more sensitive to the interest rate than the wealth of the poor.¹⁴ To see why $\varepsilon^{a_1} - \varepsilon^A > 0$ follows, note that the aggregate semi-elasticity is a weighted sum of individual elasticities:

$$\varepsilon^A = \frac{a_1}{A} \varepsilon^{a_1} + \frac{a_2}{A} \varepsilon^{a_2}.$$

Since $\varepsilon^{a_1} > \varepsilon^{a_2}$, this convex combination lies strictly below ε^{a_1} , so $\varepsilon^{a_1} > \varepsilon^A$. Because $\varepsilon^{a_1} > 0$ and $dr < 0$, the percentage decline in the wealth of the rich exceeds the percentage decline in average wealth: the general-equilibrium price effect always dampens wealth inequality.

The trickle-down effect has direct welfare implications. The interest rate decline reduces capital income (in proportion to a_i) while raising wages. Because financial income represents a higher share of the income of the rich, and labor income a higher income share of the poor, this disproportionately benefits poorer households. Proposition 3 formalizes this: the general-equilibrium price adjustment reduces welfare inequality, and under perfect competition, it increases the welfare of the poor and decreases the welfare of the rich.

Proposition 3. *Under Assumption 1, with $\beta = 0$, $\sigma > 1$, and $\Sigma < 1$, the general-equilibrium price change $dr < 0$ increases the welfare of the poor more than the welfare of the rich. Moreover, at $\mu = 1$, the price effect is fully redistributive: $\Delta U_1 < 0 < \Delta U_2$.*

Proof. See Online Appendix A.4. □

3.3 Imperfect Competition vs Trickle Down

We now examine how markups affect the trickle-down effect, proceeding in two steps. First, we show that higher markups dampen the equilibrium price response, reducing $|dr|$. Second, we show that this in turn weakens $|(\varepsilon^{a_1} - \varepsilon^A)dr|$, the overall trickle-down effect on wealth inequality.

¹⁴Note that this also accounts for the change in w , and hence the firm's parameters. Since $a(z, w(r), r)$, we have $da/dr = \partial a/\partial r + \partial a/\partial w \cdot \partial w/\partial r$.

A rise in markups operates through three channels: (i) it increases the value of firms and its sensitivity to r by raising monopoly rents; (ii) it reduces the capital stock, as the firm restricts output to protect its rents, which also lowers w ; and (iii) it raises the equilibrium interest rate, further altering K, q, Y , and w . These changes jointly affect the semi-elasticities of supply and demand, ε^A and ε^q .

To understand channel (i), note that the firm value at the steady-state can be written as

$$q = K + \underbrace{\alpha \frac{\mu - 1}{\mu} \frac{Y}{r}}_{\text{capitalized value of rents}}. \quad (15)$$

When r falls, this valuation channel increases q even if K responds only modestly, raising q/K , the average Tobin's Q —a mechanism absent under perfect competition. The valuation effect dominates, so ε^q increases with μ at a given r . The higher semi-elasticity of asset supply implies that a smaller change in r suffices to clear the asset market for a given $d \log A$, thereby weakening the trickle-down effect, as shown in Proposition 4.

This is illustrated in the right panel of Figure 6. The asset-supply curve is steeper under imperfect competition, so the equilibrium change in r is smaller for a given increase in desired savings. Moreover, the demand for physical capital differs from the firm's valuation of capital, so the equilibrium change in K is smaller than the change in q . This further dampens the change in output and wages, further weakening the trickle-down effect.

The complete proof requires working out the effects of channels (ii) and (iii) on the semi-elasticities of asset demand and supply. This is more involved and requires total differentiation of the equilibrium conditions. To obtain analytical solutions, we set $\beta = 0$ and consider a local increase in markups around perfect competition, $\mu = 1$.

Proposition 4. *For a given partial-equilibrium increase in savings $\partial \log A$, a higher level of markups μ :*

1. *Under sufficient conditions satisfied for standard calibrations, decreases the equilibrium decrease in the interest rate dr and increases the equilibrium increase in the wealth-to-output ratio $d \log(q/Y)$;*
2. *Decreases $|(\varepsilon^{a1} - \varepsilon^A)dr|$, the magnitude of the "trickle-down" effect, keeping $\varepsilon^{a1} - \varepsilon^A$ constant;*

3. Under $\beta = 0$ and locally at $\mu = 1$, and under sufficient conditions satisfied for standard calibrations, $\varepsilon^{a_1} / \varepsilon^A$ decreases with μ , further weakening the trickle-down effect.

Proof. See Online Appendix A.5 and Online Appendix A.6. □

4 Numerical Results

The previous section proved analytically that an increase in permanent labor income inequality decreases the interest rate at the steady-state. This change in prices, in turn, dampens wealth inequality. When markups are high, the semi-elasticity of asset supply is higher, decreasing the change in r , and limiting the "trickle-down" effect.

We now use our full quantitative model to answer two questions. First, what is the magnitude of the "trickle-down" effect? Second, what are the dynamics of wealth inequality following a shock? This is key, because even if, at the steady-state, markups only dampened the "trickle-down" effect, it turns out that, in a transition, they *overturn* it. Under imperfect competition, a rise in PLI creates capital gains that increase the returns for the wealthy, persistently pushing up wealth-inequality.

4.1 Calibration

The calibration targets moments from the U.S. economy in 1970 at the annual level.

Taste for wealth and preferences. We set the parameters governing the taste for wealth Σ, γ and ζ , along with the death probability ξ , to match four targets¹⁵: (i) an elasticity of consumption to PLI of 0.9; (ii) the Pareto tail of wealth ζ^a ; (iii) the relative Pareto tail of consumption to wealth ζ^c / ζ^a ; (iv) the observed top 1% wealth share in 1970.

To set those targets, we follow Gaillard et al. (2023) and derive an analytical approximation of the Pareto tails in the model, which yields¹⁶

$$\zeta^a = -\frac{\log(1 - \xi)}{\log(1 + r)}, \quad \zeta^c = \zeta^a \frac{\sigma}{\Sigma}.$$

¹⁵In Elina & Huleux (2026), we follow a similar strategy in the context of a HANK model with nominal rigidities, but targeting an aggregate MPC of 0.5 instead of a long-term elasticity of consumption to permanent labor income of 0.9.

¹⁶See Appendix A.1 for the derivation of those approximations.

Table 1: Calibration of the model

Parameter	Description	Value	Source
<i>Households</i>			
β	Discount factor	0.97	Annual r of 3%.
σ	CRRA on consumption	2.00	Standard.
Σ	CRRA on wealth	0.89	Pareto coeff. on cons./wealth.
γ	Strength of taste for wealth	0.48	Top 1%.
ζ	Stone-Geary shifter	4.12	Elasticity of Consumption to PLI.
ξ	Death probability	0.04	Pareto coeff. on cons./wealth.
ρ^e, σ^e	Corr. and var. of shocks (annual)	(0.9, 0.5)	
$\{K_0, K_1, K_2\}$	Portfolio shares	(2.54, 0.5, 5.77)	SCF (2019).
<i>Firms</i>			
α	Cobb-Douglas	0.33	Standard
δ	Depreciation	0.06	Annual W/Y 4.0
Z	TFP	0.71	$Y = 1$
μ	Markup	1.30	De Loecker et al (2020)
F	Fixed cost	0.00	$q/K = 0.7$
<i>Government</i>			
B_{ss}	Public debt	0.25	NIPA.
G_{ss}	Gov. Spendings	0.23	Residuals.
τ_{ss}	Tax rate	0.30	Standard.
λ	Tax progressivity	0.12	Standard.

We take their long-term estimates of the Pareto tails in the U.S., $\zeta^c = 3.02$ and $\zeta^a = 1.35$, and assuming a post-tax interest rate of $r = 3\%$ and a CRRA coefficient on consumption $\sigma = 2$, we obtain $\xi = 0.04$ and $\Sigma = 0.89$. We then pick $\gamma = 0.48$ and $\zeta = 4.12$ to match a top 1% wealth share of 26.9% in the U.S. in 1970, and an elasticity of consumption to permanent labor income of 0.9, as in Straub (2019). Finally, we set the discount factor β such that the market-clearing interest rate is $r = 3\%$, which yields $\beta = 0.97$.

Marginal propensity to save. Although we do not target the marginal propensity to save directly, our model reproduces its positive relationship with permanent income we observe in the data, and that we have identified as a key feature in our analytical model. Table 2 shows the MPS per different type of permanent income. Our results are in line with the MPS of 0.5 for the top 5% of the labor income distribution reported in Kumhof et al. (2015), using the SCF.

Table 2: Marginal propensity to save by permanent income group

Bottom 50%	Next 40%	Next 9%	Next 0.9%	Top 0.1%
0.09	0.31	0.38	0.49	0.56

Note: This table shows the marginal propensity to save (MPS) for each permanent income group at the initial steady state.

Labor income inequality. We set the persistence of idiosyncratic shocks ρ^ε to 0.9 and their standard deviation σ^ε to 0.5, following Straub (2019), and discretize the process using the Rouwenhorst method. We then set the permanent productivity levels z_i in order to match the observed distribution of pre-tax labor income in the U.S., as reported by Piketty et al. (2018). We use five permanent types, representing the bottom 50%, next 40%, next 9%, next 0.9%, and top 0.1% of the distribution of labor income. Since the model features single-agent households with one unit of labor, the unit of observation maps directly to the equal-split adult in the PSZ distributional national accounts.

Portfolio shares. Capital gains generated in the period $t = 0$, following a shock, are paid according to the exogenous portfolio rule defined in Equation 3. We calibrate the parameters of this sigmoid function to match the share of risky assets along the wealth distribution, using the Survey of Consumer Finance. Appendix A.3 provides details on how we calibrate the parameters of this sigmoid function.

Government. We set the government debt level to match the observed public debt-to-GDP ratio B/Y held by households in the pre-1970 era of 25%, a tax-rate τ of 30%, and compute government spendings G as the residual of the steady-state budget constraint of the

government. We follow Ferriere & Navarro (2018) and set the progressivity parameter to $\lambda = 0.12$.

Firms. We calibrate the firm side of the model by setting the markup to $\mu = 1.3$, the average markup in the U.S. between 1960 and 1980 documented in De Loecker et al. (2020). In the quantitative analysis, we subsequently assess the sensitivity of our results to a higher value of $\mu = 1.45$, the average in the U.S. during the 1960-2019 period. We set the elasticity of output with respect to capital to $\alpha = 1/3$, and calibrate Z , and δ to jointly match an output level of $Y = 1$ and a wealth-to-output ratio $(q + B)/Y = 4$.

4.2 Matching the Wealth Distribution at the Steady-State

Table 3 reports the distribution of wealth in the baseline model, together with the equilibrium interest rate and the wealth-to-output ratio, for an economy calibrated to the U.S. in 1970.¹⁷ The model closely matches the level of wealth inequality observed in the data. This result is non-trivial, as standard one-asset heterogeneous-agent models typically struggle to replicate the upper tail of the wealth distribution, including its Pareto behavior, when accumulation is driven solely by precautionary motives.

In our framework, this limitation is overcome by introducing a non-homothetic taste for wealth and a positive death probability. As households become sufficiently wealthy, the marginal utility of consumption declines faster than the marginal utility of wealth, inducing them to continue accumulating assets. In the limit, rich households save almost all of their income, and their policy function asymptotically converges to $a' = a(1 + r)$, as illustrated in Figure A.1.

The death probability plays a crucial role in preventing the wealth distribution from becoming degenerate. By periodically removing households from the economy and redistributing their assets, it continuously replenishes the lower part of the wealth distribution. In an infinite-horizon setting, this mechanism can be interpreted as capturing disaster risk or the stochastic termination of dynasties.

This mechanism can be seen in Table 4, which decompose the factors behind the distribution of wealth in the model: removing the taste for wealth decreases the top 1% wealth

¹⁷Online Appendix C describes the numerical solution method used to solve the steady state and transition dynamics of the model.

Table 3: Distribution of wealth in the baseline model for the year 1970

Name	Bottom 90%	Top 10%	Top 1%	Top 0.1%	Top 0.01%	r	A/Y
Model	31.7	68.3	27.3	11.0	4.8	3.0	4.0
Data PSZ	30.9	69.1	26.9	9.5	3.3		

Note: This table shows the distribution of wealth in the baseline model, along with the equilibrium interest rate and wealth-to-output ratio, in levels. For example, the column 'T-10%' for the baseline row means that the top 10% wealth share was 68.3%.

share by 14.5 p.p. and the top 0.1% wealth share by 8.5 p.p.¹⁸ The remaining main drivers of inequality at the steady state are the distribution of permanent labor income (line 2), the imperfect competition setting (line 3), and the proportionality of bequest (line 4). Note that because of the non-linear nature of the model, the different effects do not sum to the total wealth shares.

Table 4: Contribution of different channels to steady state wealth shares

#	Scenario	Bottom 90%	Top 10%	Top 1%	Top 0.1%	Top 0.01%	r	A/Y
1	Homothetic prefs. ($\gamma = 0$)	17.7	-17.7	-15.2	-8.7	-4.3	0.0	-0.0
2	Perfect competition ($\mu = 1$)	4.6	-4.6	-6.5	-4.8	-3.1	-0.9	-97.9
3	No prop. inheritance	-0.4	0.4	1.5	2.4	2.6	0.4	-30.3
4	No trans. shocks ($\sigma_\varepsilon = 0$)	-0.1	0.1	3.7	3.4	2.5	0.2	-17.3
5	No redist. ($\lambda = 0$)	-1.4	1.4	0.8	-0.0	-0.4	-0.2	15.8

Note: This table displays the p.p. change in the wealth distribution, the interest rate, and the wealth-to-output ratio, for each quantile after removing a given feature of the model, keeping the calibration constant. For example, the first row says that setting homothetic preferences ($\gamma = 0$) decreases the top 1% wealth share by 15.2 p.p.

Removing imperfect competition lowers wealth inequality by increasing the stock of physical capital and aggregate output, reducing the interest rate on wealth, and raising wages. The resulting decline in r disproportionately affects wealthy households, whereas the increase in w benefits the lower part of the distribution, for whom labor income represents a larger share of total income.

Finally, note that idiosyncratic productivity plays a small role in defining the wealth distribution in our model. As pointed out by Hubmer et al. (2021), although idiosyncratic productivity shocks are a source of wealth inequality (and what makes the wealth-distribution smooth in our model), decreasing the variance of idiosyncratic shocks actu-

¹⁸In this scenario, we recalibrate the discount factor β to keep r at 3% and the wealth-to-output ratio at 4. Keeping the discount factor β at its calibrated value with a taste for wealth would imply a highly counterfactual interest rate of around 100%, if such a steady-state existed. For the remaining rows, we keep β at its baseline value, and let r adjust to clear the asset market.

ally increases wealth inequality by reducing precautionary savings at the bottom of the distribution.

4.3 Steady-state changes

To consider how the overall change in prices due to an increase in permanent labor income inequality varies with μ , we compare three cases: (i) our benchmark economy, with $\mu = 1.3$, (ii) a high-markup economy, with $\mu = 1.45$, and (iii) a perfect competition economy, with $\mu = 1$. For each of those cases, we solve two steady-states: one associated with low permanent labor income inequality, calibrated on the distribution of labor income in the U.S. in 1970, and one with high permanent labor income inequality, calibrated on the distribution of permanent labor income inequality in 2019.

Table 5 reports the relevant elasticities and the implied change in r across the three economies. Consistent with the mechanism highlighted in Section 3, the asset-supply elasticities are higher in economies with higher markups. In the perfect competition economy, the interest rate decreases by 105 b.p., while wages and output increase by 6%. In our benchmark economy with a markup of 1.3, the interest rate decreases by only 70 b.p., and wages and output increase by 4.1%.¹⁹ The wealth-to-output ratio increases by 70 p.p. in our benchmark model, against 32 p.p. in the perfect competition economy. Increasing markups from 1.3 to 1.45 further reduces the change in r to 65 b.p., and the increase in the wealth-to-output ratio to 79 p.p.

Table 5: Elasticities at the steady-state

Model	$\partial \log A$	ε^q	ε^A	dr	$d \log w$	$d \log Y$	$d(A/Y)$
$\mu = 1$	0.5	15.0	25.0	-105.4	6.3	6.3	31.7
$\mu = 1.3$	0.4	25.4	25.2	-70.2	4.1	4.1	70.0
$\mu = 1.45$	0.4	27.6	25.3	-64.8	3.7	3.7	78.8

Note: Interest rate change dr reported in basis points (1 p.p. = 100 basis points). Note that the reported change dr does not fully match our baseline formula because of non-linearities in the model, and the large size of the shock.

These price changes dampen the increase in wealth inequality provoked by the rise in permanent labor income inequality. To quantify this channel, Table 6 decomposes the

¹⁹Our estimated impact of labor income inequality on the interest rate and the wealth-to-output ratio, are also in line with the estimated impact in Auclert et al. (2025).

general-equilibrium contribution of r , w , and η to changes in average wealth levels within each quantile, holding permanent labor income constant and varying only equilibrium prices. Because permanent labor income is more equally distributed than capital income, and capital income represents a larger share of total income at the top of the wealth distribution, the decline in r reduces income disproportionately at the top, while the rise in w benefits households more evenly. The general-equilibrium price adjustment thus dampens wealth inequality. The total general-equilibrium effect decreases the wealth levels of the top 1% (0.1%) by 33% (50%) under perfect competition, compared with 27% (43%) in our benchmark model with markups. Most of the decrease in top wealth levels comes from the decline in the real interest rate. Table B.2 in the Appendix shows that those results carry over to wealth shares as well.

Table 6: General equilibrium effect on wealth levels at the steady-state

Variable	Model	Bottom 90%	Top 10%	Top 1%	Top 0.1%	Top 0.01%	Average
r	$\mu = 1$	-25	-32	-42	-57	-75	-30
	$\mu = 1.3$	-18	-28	-38	-52	-68	-25
	$\mu = 1.45$	-18	-27	-37	-51	-67	-24
w	$\mu = 1$	14	11	11	12	13	12
	$\mu = 1.3$	7	7	7	7	8	7
	$\mu = 1.45$	6	6	6	6	7	6
η	$\mu = 1$	9	7	8	9	10	8
	$\mu = 1.3$	14	12	13	15	17	13
	$\mu = 1.45$	14	13	14	16	18	14
Total	$\mu = 1$	-6	-20	-33	-50	-71	-16
	$\mu = 1.3$	0	-15	-27	-43	-62	-10
	$\mu = 1.45$	1	-14	-26	-42	-60	-9

Note: This table displays the perc. change in the average wealth levels (divided by average income) for each quantile due to the general equilibrium effect. Note that due to non-linearities in the policy functions, the sum of the partial effects do not add up to the total general equilibrium effect.

This section thus confirms that our previous theoretical findings are economically significant. The general-equilibrium effect dampens wealth inequality by reducing the interest rate and increasing the wage, but the magnitude of this effect decreases strongly when markups are present.

4.4 One-Time Permanent Transitions

We now analyze the transitional dynamics that follow a one-time increase in permanent labor income inequality. Figure 7 illustrates the joint evolution of the wealth-to-output ratio, average Tobin's Q, and the interest rate along the transition. A sharp contrast emerges between economies with perfect competition and those with imperfect competition.

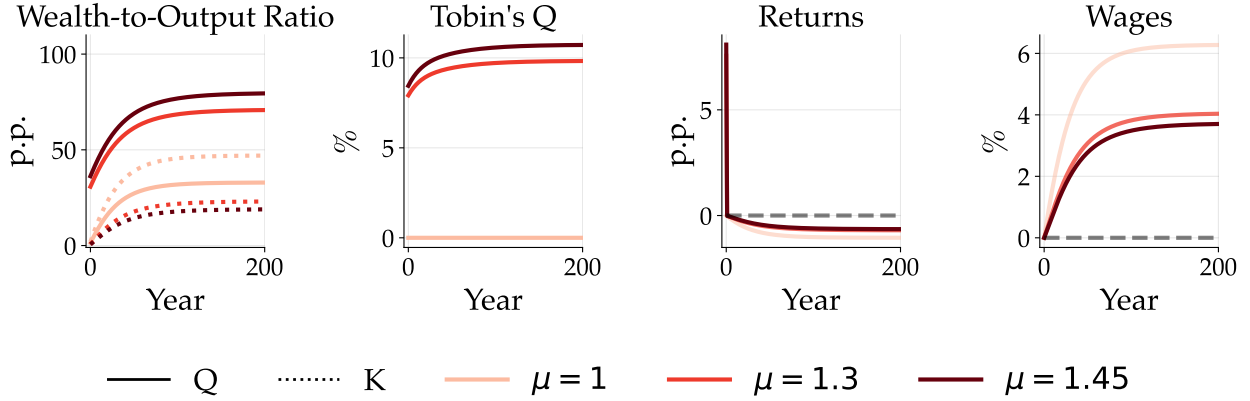


Figure 7: Change in wealth-to-output, average Tobin's Q and interest rate, following a one-time increase in permanent labor income inequality

Note: The interest rate change in the third panel accounts for capital gains in the first period, and is computed as $r^{full} = r + cap_0 \times \int s(a) d\Phi / A$. For visibility, we do not show the return on capital gains directly (since $\int s(a) d\Phi < A$), which is higher in the first period than what is plotted here.

Under perfect competition, the rise in aggregate savings induced by higher permanent labor income inequality is absorbed through an expansion of the physical capital stock. Asset prices follow the stock of capital one-for-one, so average Tobin's Q is constant and the adjustment operates entirely through quantities. The interest rate declines smoothly as capital accumulates, and the wealth-to-output ratio increases only gradually. Because wealth accumulation reflects slow movements in physical assets rather than valuation effects, its dynamics are slow and progressive.

When firms operate under imperfect competition, the long-run decline in the real interest rate translates primarily into higher firm valuations rather than a commensurate increase in the capital stock. Since equity prices are forward looking, the higher long-run value of firms induces an immediate jump in their price at the time of the shock. This generates a jump in Tobin's Q and in the wealth-to-output ratio on impact, together with a temporary rise in the realized return due to unexpected capital gains. This pattern — an increase in the wealth-to-output ratio driven mostly by valuation effects — matches the dynamics

observed in the U.S. data that we documented previously.

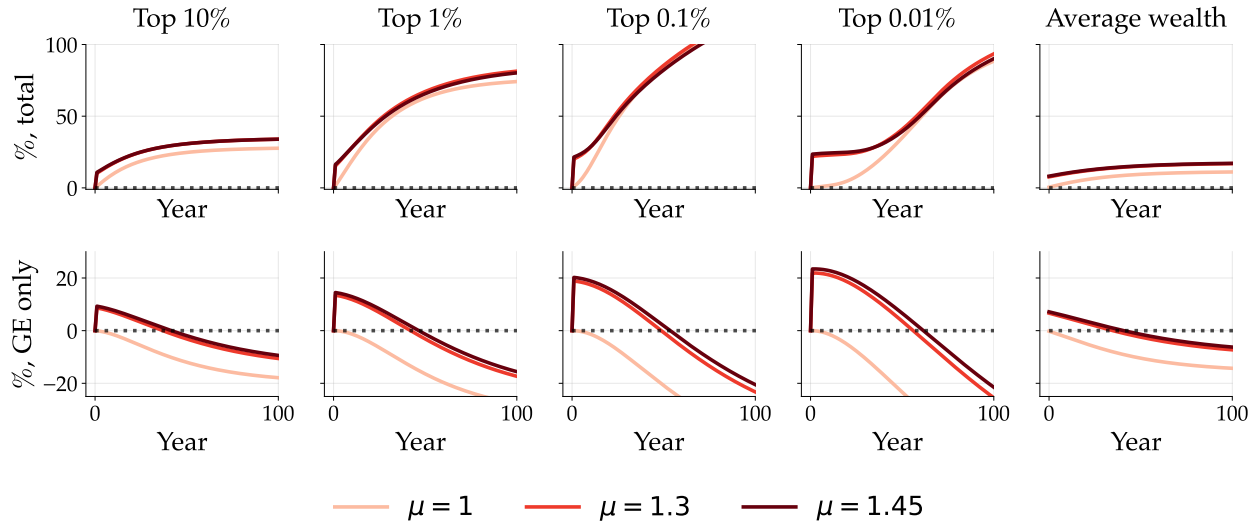


Figure 8: Change in the top wealth levels following a one-time increase in permanent labor income inequality

Note: The top row shows the p.p. change in the top wealth levels following a permanent labor income inequality shock. The second row shows the share of this change explained by the general-equilibrium effect.

Figure 8 reports the total change in the top wealth levels (first row) and the general-equilibrium effect (second row). Under perfect competition, this contribution is strictly negative: the decline in interest rates reduces financial income at the top, while labor income is less unequally distributed than capital income, and represents a larger share of total income at the bottom. An increase in labor income hence dampens wealth concentration. With imperfect competition, however, the valuation channel dominates in the short and medium run.

The price jump at the time of the shock generates sizable capital gains for asset-rich households, producing an immediate increase in top wealth levels and accelerating the convergence toward the high-inequality steady state. In this case, the general-equilibrium effect is positive on impact and quantitatively large in the early phases of the transition. The general-equilibrium effect increases the average wealth of the top 1% (0.1%) by 18% (20%) on impact, and the effect remains positive for more than 50 years.

Appendix Figure B.1 decomposes this GE effect between a change in r (without capital gains), a change in w , a change in capital gains, and the change in bequests η . Capital gains at the time of the shock account for the bulk of the positive contribution, while changes in the long-run interest rate, wages, and bequests play secondary roles in the short run. In

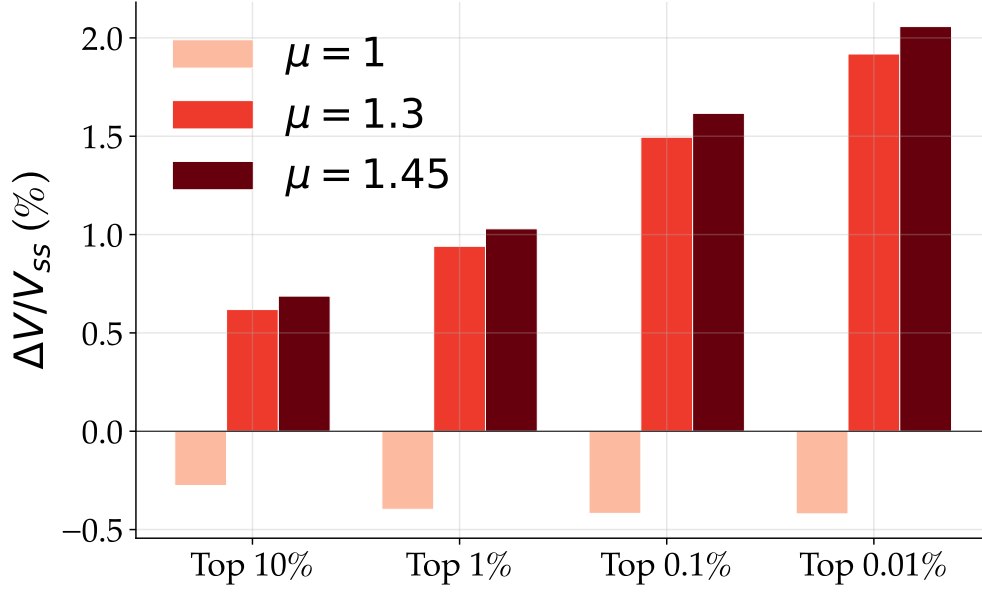


Figure 9: Percentage change in welfare at $t = 0$ induced by the general-equilibrium effect, following a one-time increase in permanent labor income inequality

the long run, the decrease in the interest rate starts to hurt wealthy households, and the GE effect turns negative, in line with our steady-state analysis. Overall, the results highlight that imperfect competition transforms higher desired savings into valuation effects rather than quantities. Capital gains front-load wealth inequality and substantially alter the speed and nature of the transition.

In our model, this change in prices has strong welfare implications. Figure 9 shows the percentage change in welfare at $t = 0$ induced by the general-equilibrium effect, following a one-time increase in permanent labor income inequality. Under perfect competition, richer households are hurt by the endogenous change in prices, while they benefit from it under imperfect competition. In both cases, the magnitude of this effect increases along the distribution of wealth.

4.5 Gradual Transition: Myopic Expectations

The previous subsection considered a stylized experiment in which permanent labor income inequality jumps immediately from its 1970 level to its 2019 level. In the data, however, inequality rose gradually over roughly five decades, so the associated equilibrium adjustments plausibly unfolded over time. To measure more precisely the cumulative general-equilibrium mechanism identified above, we therefore compute a 50-period tran-

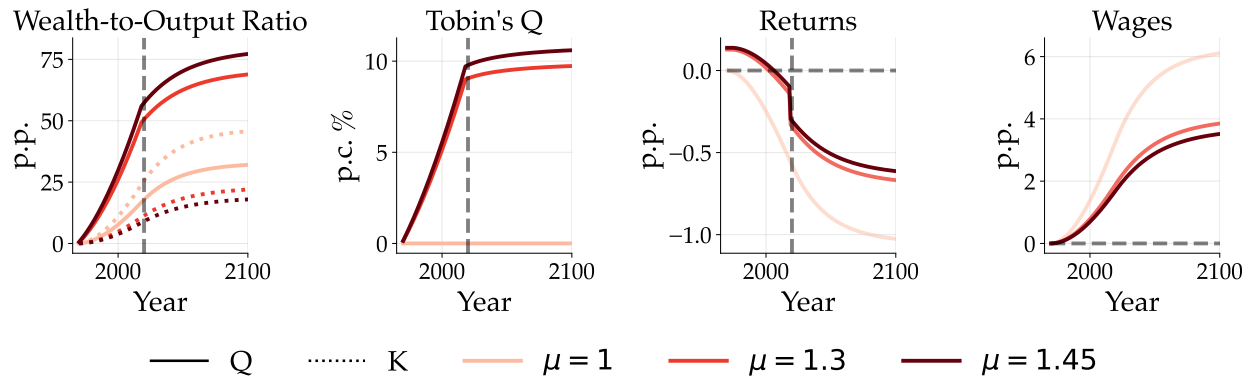


Figure 10: Change in the wealth-to-output ratio, Tobin's Q, returns, and wages, following a progressive increase in permanent labor income inequality

sition that matches the observed progressive increase in permanent labor income inequality.

We model this transition as myopic: in each period, the economy is hit by a one-time, unanticipated MIT shock to inequality, and households do not foresee future shocks. Conditional on the shock realized in a given period, however, households correctly understand its implications for current and future equilibrium prices and make optimal decisions given those anticipated price paths. This assumption is clearly a strong simplification, but it captures—in reduced form—a class of behavioral frictions emphasized in recent heterogeneous-agent work. In particular, it is consistent with the idea that households infrequently update their information about aggregate conditions and asset prices, as in Auclert et al. (2021).²⁰

Figure 10 reports the change in the wealth-to-output ratio, the Tobin's Q, the interest rate (accounting for capital gains), and wages, in our myopic transition. We find a similar pattern in this exercise as in our one-time transition. The wealth-to-output ratio increases more and faster when markups are higher. This change is driven mostly by an increase in the value of the firms, instead of their stock of physical capital, pushing up the average Tobin's Q. The interest rate r decreases in the long run, but is temporarily pushed up by capital gains, when markups are high.

Figure 11 reports the change in top wealth levels in the model. The GE effect is positive,

²⁰A natural alternative would be to assume perfect foresight: households in 1970 would correctly anticipate the entire subsequent rise in permanent labor income inequality and the associated evolution of returns. We view this as implausible, since it requires an accurate forecast of a long-run distributional trend that was not well understood even by economists until much later.

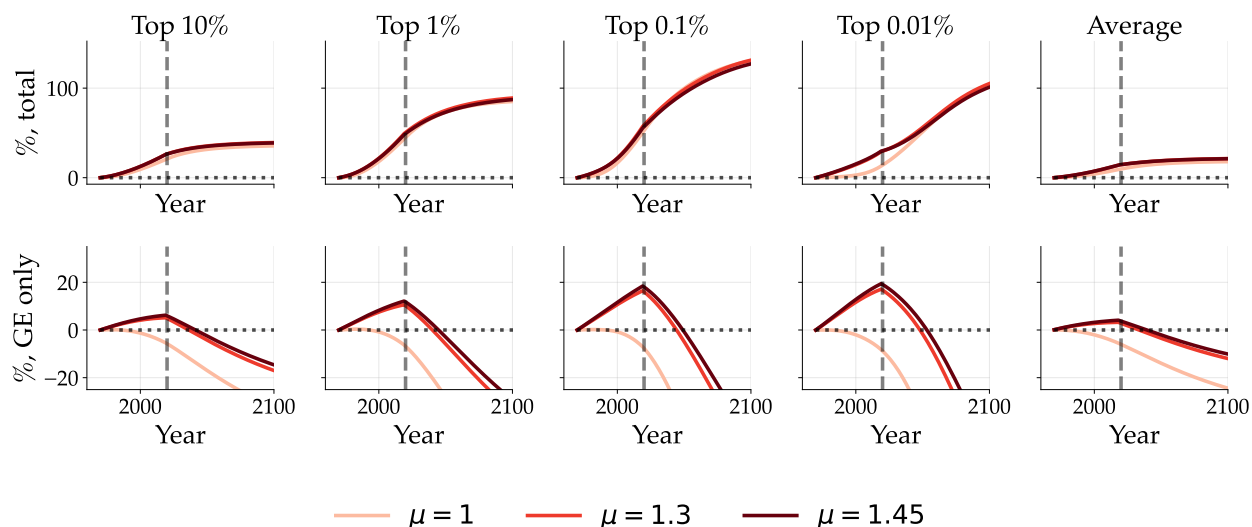


Figure 11: Change in the top wealth levels following a progressive increase in permanent labor income inequality

and increased the average wealth of the top 1% by 10% and the top 0.1% by 16% in 2020, against 3% for an average household in the economy, and remains positive for the next 25 years. Thus, our model predicts that even in the absence of an additional rise in permanent labor income inequality, wealth inequality will keep rising in the future, driven partly by the persistent effect of capital gains. This shows that although the "trickle-down" mechanism identified in Section 4.3 is still present in a world with imperfect competition, its effects might take decades, if not more, to materialize. In the short and in the medium run, wealth inequality is still pushed up by the large increase in the financial value of firms, driven by lower interest rates.

Figure 12 reports the change in welfare induced by the general-equilibrium effect. Even if the magnitude of the welfare change is smaller, as the price changes are smoothed over time and hence discounted at $t = 0$, the same pattern remains: the presence of markups changes the sign from negative to positive and benefits mostly richer households. The key is the relative timing of the two channels: the valuation effect generates capital gains that disproportionately benefit wealthy households in the short and medium run, whereas the "trickle-down" channel operates only in the very long run. Because those long-run income changes are heavily discounted, their present-value impact on welfare at $t = 0$ is small relative to the front-loaded capital gains.

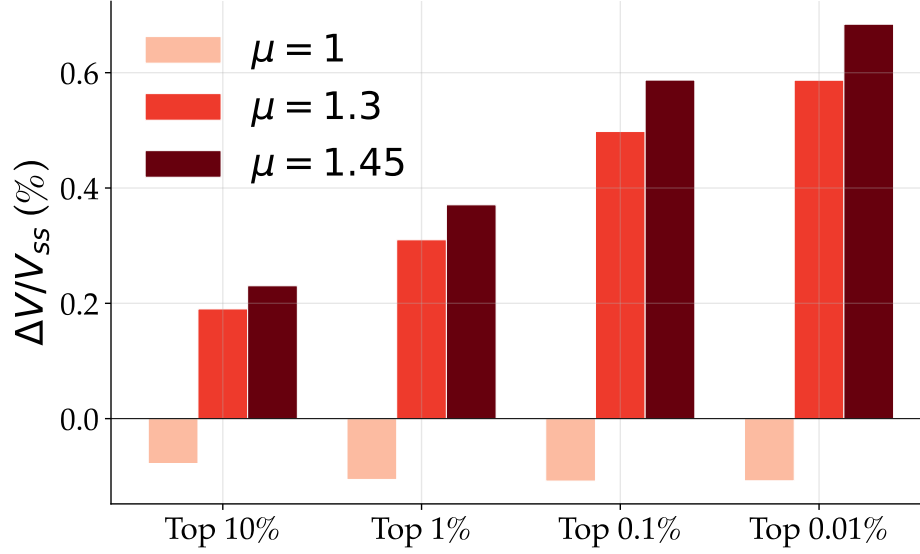


Figure 12: Change in welfare induced by the GE effect, following an increase in permanent labor income inequality

5 Conclusion

This paper shows that imperfect competition fundamentally alters the general equilibrium transmission of rising permanent labor income inequality to wealth inequality. With non-homothetic preferences on wealth, higher permanent labor income inequality raises desired savings and lowers the interest rate. Under perfect competition, this generates a “trickle-down” effect—lower returns and higher wages—that dampens wealth concentration, reducing top 1% (0.1%) wealth levels by 33% (50%) at the steady state.

Markups weaken this mechanism: the steeper asset-supply curve implies a smaller decline in r , and the resulting increase in firm valuations is absorbed through higher equity prices rather than physical capital accumulation. In our calibrated economy, the trickle-down effect shrinks, decreasing the average wealth level within the top 0.1% by 43%. Along the transition, the valuation channel dominates in the short and medium run. In our main quantitative exercise, matching the observed rise in U.S. permanent labor income inequality from 1970 to 2019, the model predicts that capital gains increased the average wealth level within the top 1% (0.1%) by 8.4% (13.4%). This effect will remain positive in the coming decades, even in the absence of further rise in permanent labor income inequality.

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A Additional Details on Calibration

A.1 Deriving the Pareto tail in a Heterogeneous-agent model

The proof draws heavily on Gaillard et al. (2023). We derive the asymptotic right tails of the wealth and the consumption distributions in a simpler framework without scale and type dependence.

The problem of the household writes

$$V(a_t, z_t) = \frac{c_t^{1-\sigma}}{1-\sigma} + \gamma \frac{(a_{t+1} + \zeta)^{1-\Sigma}}{1-\Sigma} + \beta(1-\xi)\mathbb{E}[V(a_{t+1}, z_{t+1})]$$

subject to

$$a_{t+1} + c_t = a_t(1+r) + z_t w$$

and a zero borrowing constraint

$$a_{t+1} \geq 0.$$

Households are subject to persistent idiosyncratic productivity shocks z_t and to random death shocks arriving with probability ξ . The Euler equation of the household is

$$c_t^{-\sigma} = (1-\xi)\beta(1+r)\mathbb{E}[c_{t+1}^{-\sigma}] + \frac{\gamma}{A} (a_{t+1} + \zeta)^{-\Sigma}.$$

The key thing to note is that, when $a \rightarrow \infty$, labor income $z_t w$ becomes negligible and we can abstract from uncertainty coming from idiosyncratic productivity shocks. The problem now writes

$$V(a_t) = \frac{c_t^{1-\sigma}}{1-\sigma} + \gamma \frac{(a_{t+1} + \zeta)^{1-\Sigma}}{1-\Sigma} + \beta(1-\xi)V(a_{t+1})$$

subject to

$$c_t + a_{t+1} = a_t(1+r).$$

The Euler equation now writes

$$c_t^{-\sigma} = (1-\xi)\beta(1+r)c_{t+1}^{-\sigma} + \gamma (a_{t+1} + \zeta)^{-\Sigma}.$$

where $c_t + a_{t+1} = a_t(1 + r)$. Guess that $c = Ba^{\Sigma/\sigma}$, this implies that

$$a' = a(1 + r) - c = a(1 + r) - Ba^{\Sigma/\sigma}.$$

This implies that as $a \rightarrow \infty$, $a' \sim a(1 + r)$. We use a guess and verify technique. As $a \rightarrow \infty$, we have

$$c' = C(a')^{\Sigma/\sigma} = B(a(1 + r) - Ba^{\Sigma/\sigma})^{\Sigma/\sigma} = B(a(1 + r))^{\Sigma/\sigma}$$

$$c^{-\sigma} = (1 - \xi)\beta(1 + r)B^{-\sigma}(a(1 + r))^{-\Sigma} + \gamma a^{-\Sigma}.$$

$$c = [a(1 + r)]^{\Sigma/\sigma} \left[(1 - \xi)\beta(1 + r)B + \gamma \right]^{-1/\sigma}.$$

Which verifies our guess. To find B , solve

$$B = \left[(1 - \xi)\beta(1 + r)B^{-\sigma} + \gamma \right]^{-1/\sigma}$$

with a numerical solver. Adjusting for the risk probability, we have $a' = (1 - \xi)(1 + r)a = G'a$.

We can now derive the Pareto tail in this case. Beare and Toda (2022) show that the stationary wealth distribution has a Pareto upper tail where the moment generating function is

$$M_{ss'}(z) = \mathbb{E}[e^{z \log G'} | s_t = s, s_{t+1} = s'] = \mathbb{E}[G'^z | s_t = s, s_{t+1} = s'].$$

In our case, define $s = 0$ an alive household and $s = 1$ a dead household. We have

$$G_{s=0} = (1 + r), G_{s=1} = 0$$

The matrix $M(z) = (M_{ss'}(z))$ thus writes

$$M(z) = \begin{bmatrix} (1 + r)^z & 0 \\ (1 + r)^z & 0 \end{bmatrix}$$

And the Pareto parameter ζ is then defined by

$$\rho(P \circ M(z)) = 1$$

where $\circ$ is the Hadamard (entrywise) product and ρ the spectral radius. We have

$$P \circ M(z) = \begin{bmatrix} (1 - \xi)(1 + r)^z & 0 \\ (1 - \xi)(1 + r)^z & 0 \end{bmatrix}$$

And we have $\rho(P \circ M(z)) = (1 - \xi)(1 + r)^z = 1$. Solving for z yields

$$z = \zeta = \frac{-\log(1 - \xi)}{\log(1 + r)}$$

which is the Pareto tail.

To obtain the Pareto tail for consumption, note that $c = Ba^{\Sigma/\sigma}$ implies that $a = c^{\sigma/\Sigma} B^{-\sigma/\Sigma}$.

We now that a follows a Pareto distribution with tail ζ . Thus,

$$P(c > x) = P(Ba^{\Sigma/\sigma} > x) = P\left(a > \frac{x^{\sigma/\Sigma}}{B}\right)$$

Since we know that $P(a > y) \sim Cx^{-\zeta}$, we have $P(c > y) \sim C((y/B)^{\sigma/\Sigma})^{-\zeta}$. This means that the distribution for consumption follows a Pareto distribution with tail

$$\zeta_c = \zeta\sigma/\Sigma.$$

Now, Gaillard et al (2025) show that in the U.S., $\zeta_c/\zeta \approx 2.24$. Setting $\sigma = 2$, this implies $\Sigma = 0.89$.

Assuming an interest rate of 5%, and targeting a Pareto tail for wealth of 1.23, this yields a probability to die of

$$\xi = 1 - \exp(-\zeta \log(1 + r)) \approx 0.058.$$

A.2 Unit of observation: equal-split adjustment of the SCF

The Distributional National Accounts of Piketty et al. (2018) report inequality statistics per *equal-split adult*: for married couples, total household income and wealth are divided equally between the two spouses, and each spouse is treated as a separate observation.

The Survey of Consumer Finances (SCF), by contrast, uses the household—or more precisely, the *primary economic unit* (PEU)—as its unit of observation. All income and wealth

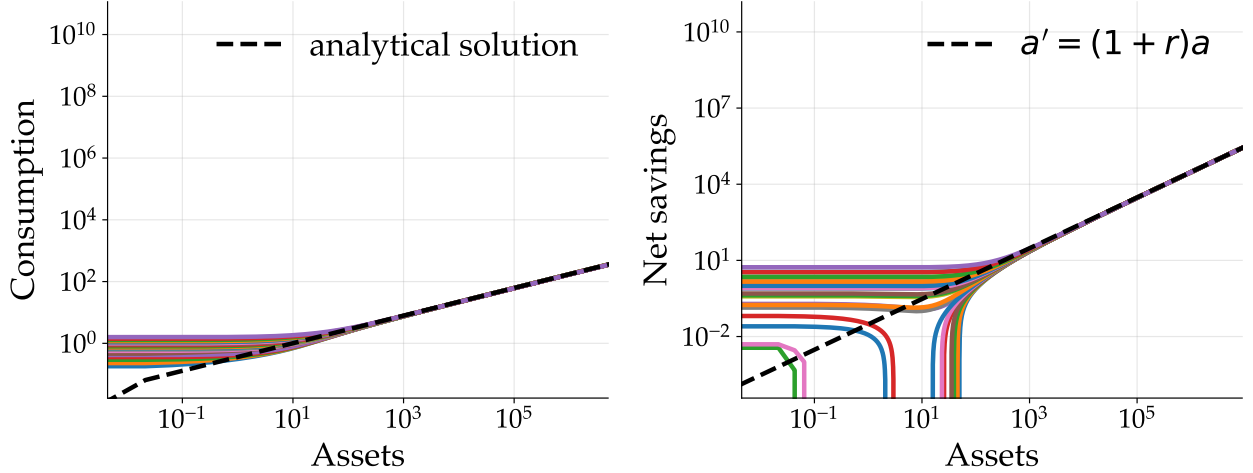


Figure A.1: Policy function for savings and consumption at the steady state

Note: The first panel shows the consumption function, as a function of wealth (x-axis), and idiosyncratic types (different colors), versus our analytical solutions for high levels of wealth. The second panel plots the policy function for the change in wealth ($a' - a$) versus the approximation $a' = a(1 + r)$.

variables are reported at the household level, without distinguishing between spouses.

To ensure comparability between the two data sources, we convert the SCF to a per-adult basis by applying an equal-split adjustment: for each married or partnered household (identified by the `MARRIED` variable in the SCF), we divide all dollar-denominated variables by two and multiply the survey weight by two. Single-person households are left unchanged. This transformation ensures that top shares and portfolio compositions computed from the SCF are measured on the same per-adult basis as the PSZ distributional data and as the agents in our model.

A.3 Calibration of the exogenous portfolio share

We follow Auclert & Rognlie (2020) and calibrate our portfolio share function on data from the Survey of Consumer Finance, using all available years from 1989 to 2019. We define the broad share measure as a residual:

$$\text{broad share} = \frac{\text{NETWORTH} - \text{LIQ} - \text{SAVBND} - \text{BOND} - \text{IRAKH}}{\text{NETWORTH}}.$$

For the narrow share, we compute

$$\text{narrow share} = \frac{\text{DEQ} + \text{BUS}}{\text{NETWORTH}}.$$

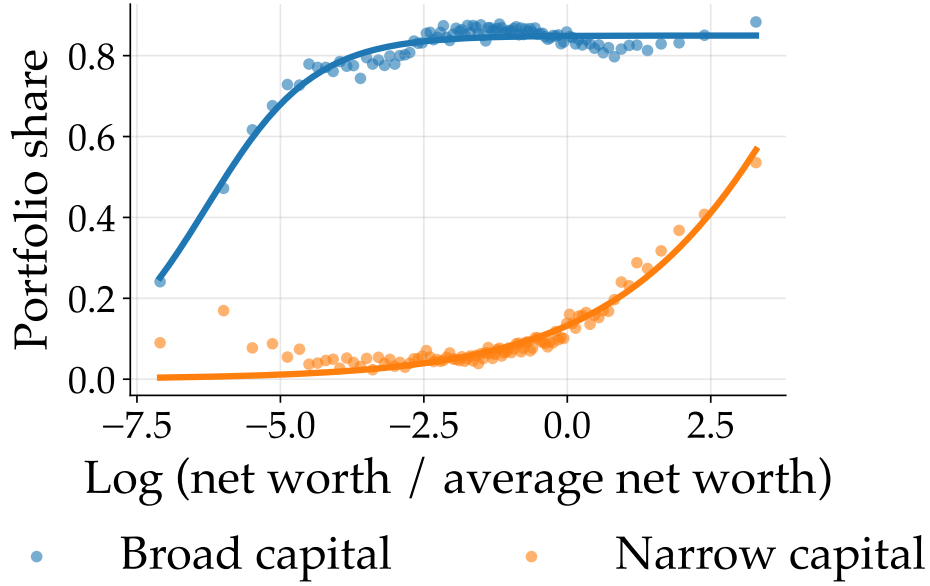


Figure A.2: Portfolio shares in the SCF and in the model

We then fit a sigmoid function over the computed calibration shares.

A.4 Markups over Time in De Loecker et al (2020)

Figure A.3 reports the evolution of markups in the U.S., as computed by De Loecker et al. (2020). The average between 1970 and 2017 is 1.38, while the average between 1960 and 1980 is 1.3. Our baseline measure uses the lower-end of this measure (1.3).

B Quantitative Model

B.1 Additional graphs and results at the steady-state and one-time transitions

This section reports additional results for the change our quantitative model. Table B.1 reports the change in the wealth levels (divided by total average income in the economy) of the top percentiles of the distribution. As mentioned previously, we see that the change in r decreases the average wealth held by households at the top of the distribution by a significant amount, while the increase in w and η increases the wealth across the distribu-

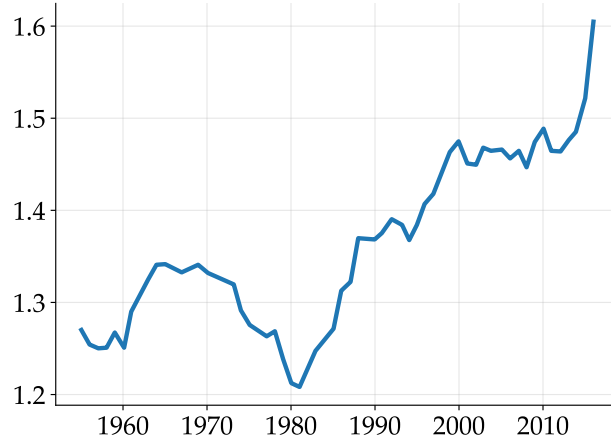


Figure A.3: Average markups in the U.S. (Figure 1 from De Loecker et al (2020))

tion. However, this GE effect varies a lot with markups μ , and is significantly dampened when markups are high.

Table B.1: General equilibrium effect on wealth levels at the steady-state

Model	Bottom 90%	Top 10%	Top 1%	Top 0.1%	Top 0.01%	Average
$\mu = 1$	-5.8	-20.2	-32.9	-50.0	-70.8	-15.8
$\mu = 1.3$	0.2	-15.1	-27.2	-43.4	-61.8	-10.2
$\mu = 1.45$	0.9	-14.3	-26.4	-42.3	-60.2	-9.4

Note: This table displays the perc. change in the average wealth levels (divided by average income) for each quantile due to the general equilibrium effect. Note that due to non-linearities in the policy functions, the sum of the partial effects do not add up to the total general equilibrium effect.

Figure B.2 shows the distribution and composition of the income distribution during the first period of a one-time shock, in the baseline model.

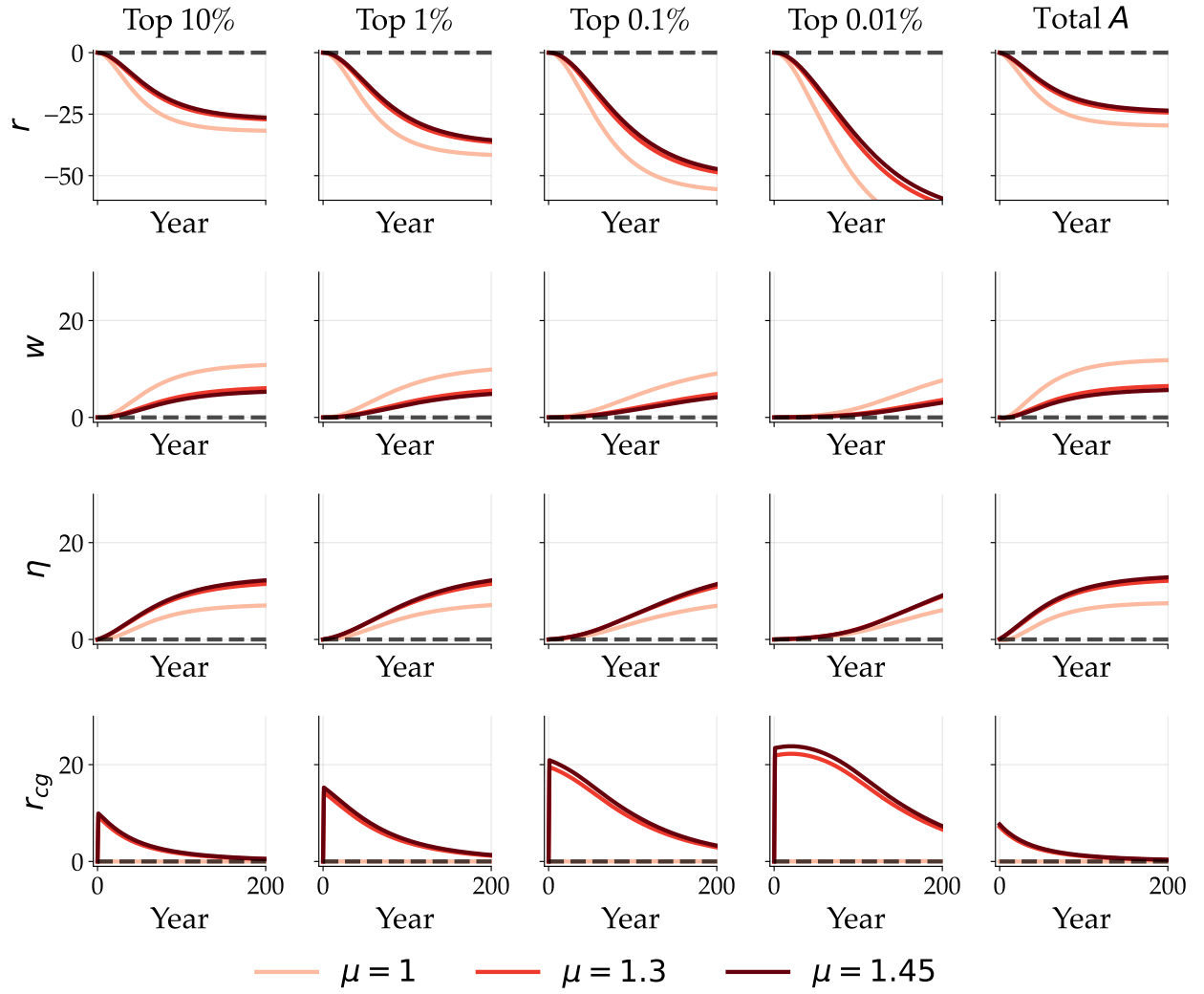
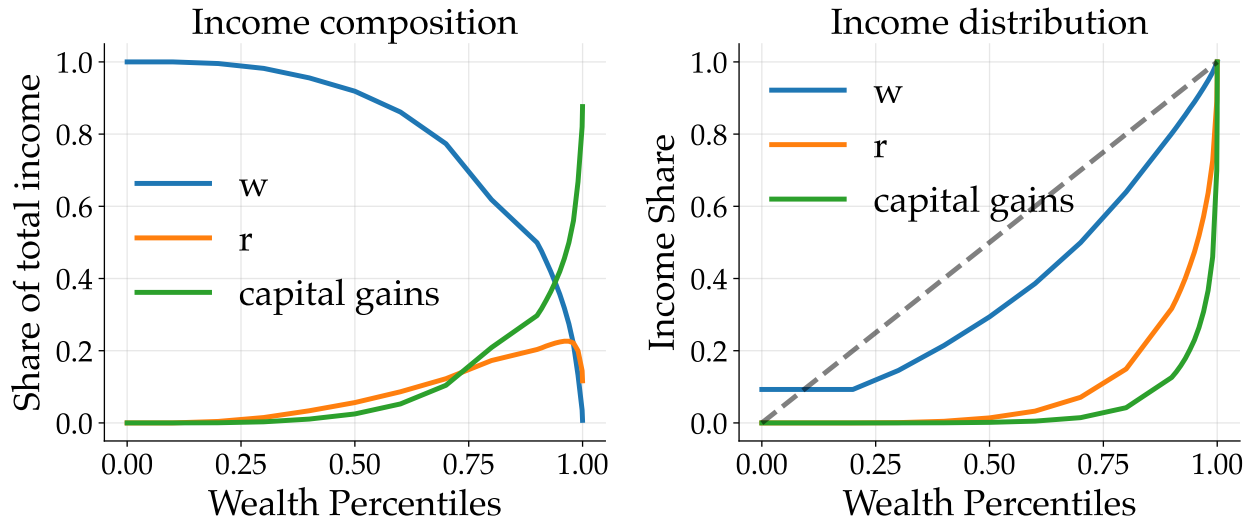


Figure B.1: Decomposition of the GE effect on the per-quantile average wealth levels.

Table B.2: General equilibrium effect on wealth shares at the steady-state

Variable	Model	Bottom 90%	Top 10%	Top 1%	Top 0.1%	Top 0.01%
r	$\mu = 1$	2.2	-2.2	-4.8	-4.1	-2.8
	$\mu = 1.3$	2.7	-2.7	-4.8	-4.0	-2.7
	$\mu = 1.45$	2.8	-2.8	-4.8	-4.0	-2.7
w	$\mu = 1$	0.5	-0.5	-0.3	-0.0	0.0
	$\mu = 1.3$	0.2	-0.2	-0.1	0.0	0.0
	$\mu = 1.45$	0.1	-0.1	-0.0	0.0	0.0
η	$\mu = 1$	0.2	-0.2	0.0	0.1	0.1
	$\mu = 1.3$	0.2	-0.2	0.1	0.2	0.2
	$\mu = 1.45$	0.2	-0.2	0.2	0.2	0.2
Total	$\mu = 1$	3.6	-3.6	-5.6	-4.3	-2.8
	$\mu = 1.3$	3.7	-3.7	-5.2	-4.1	-2.7
	$\mu = 1.45$	3.7	-3.7	-5.1	-4.0	-2.7

Note: This table displays the p.p. change in the wealth shares for each quantile due to the general equilibrium effect. Note that due to non-linearities in the policy functions, the sum of the partial effects do not add up to the total general equilibrium effect.

**Figure B.2:** Income composition and distribution along the distribution of wealth, at period $t = 0$ of a one-time permanent labor income inequality shock